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Business regulation, rule of law and formal entrepreneurship: evidence from developing countries

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Abstract

Purpose – The purpose of this paper is to examine the influence of formal rules such as business regulation and rule of law on the level of formal entrepreneurship in Latin America countries over time.

Design/methodology/approach – The authors use the panel regression techniques to examine the influence of business regulation and rule of law on formal entrepreneurship. In particular, they implement together two alternative views of formal entrepreneurship suggested in the literature: the "dual" and "legalistic" view. At the empirical level, the "dual" definition corresponds to the business owners' rate. As for the "legal" definition, it corresponds to the business registration rate. The evidence presented is based on two panels. The first panel covers the period 2004–2015, and the data set contains 180 observations in 18 countries. The second panel covers the period 2006–2015, and the data set contains 134 observations in 14 countries.

Findings – The impact of institutional variables on formal entrepreneurship depends on the definition employed: "dual" or "legal." Thus, the results suggest that business owners' rate is more substantial in Latin American countries that have weak property rights. Conversely, from a legal definition, the business registration rate is more significant in Latin American countries that have most secure property rights and fewer labor regulations. These contradictory results suggest that the legal definition of formal entrepreneurship, but not the dual definition, seems to be associated with the type of entrepreneurial activity that promotes economic growth and development.

Research limitations/implications – The results support the importance of conducting analyzes that take into account the different types of entrepreneurial activities that are present in an economy, and in addition the relevance of understanding what each measure is capturing of the heterogeneous phenomenon of entrepreneurship.

Practical implications – The results suggest that the entrepreneurship policy should focus on the quality of entrepreneurship, rather than merely seeking to increase the number of new businesses. Additionally, the results suggest that the legalistic definition of formal entrepreneurship is probably the most relevant for many policy issues.

Originality/value – The paper analyzes together two alternative views of formal entrepreneurship suggested in the literature: the "dual" and "legalistic" views. Also, the paper has used the Latinobarómetro data set, which has not been extensively used by scholars in the field of entrepreneurship and which could be useful for longitudinal research on entrepreneurial activity in Latin American countries.

Keywords Latin America, Panel data, Rule of law, Business regulation, Formal entrepreneurship

Paper type Research paper

1. Introduction

Institutional arrangements that arise in society can lead to different economic outcomes and types of entrepreneurship. As argued by North (1990), institutions, alongside the ordinary constraints of economic theory, determine the productive opportunities in society. In the same vein, Baumol (1990) argues that an institutional environment can either positively or negatively impact entrepreneurial decision making, including decisions regarding what

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type of entrepreneurship to deploy. For instance, the allocation of entrepreneurial talent toward productive activities will largely depend on the incentives and payments offered by the institutional matrix (Baumol, 1990). For Acemoglu *et al.* (2005), many societies in developing countries are trapped in a kind of extractive institutional structure that does not generate incentives nor allow its inhabitants to fully participate in the type of entrepreneurial opportunities that flourish in developed societies under inclusive institutions. For these reasons, an analysis of the kind of incentives or game rules that promote or deter the allocation of entrepreneurial talent to productive activities such as formal entrepreneurship is a field of enormous relevance, especially for developing countries (Naudé, 2010; Desai, 2011).

In recent decades, there is more considerable agreement regarding the relevance of the entrepreneurial factor to economic processes for its contribution to the creation of jobs, and the introduction of innovation and knowledge (Audretsch *et al.*, 2006). However, the empirical evidence has produced contradictory results concerning the contribution of entrepreneurship to economic development. For example, Van Stel *et al.* (2005) found that the impact of entrepreneurial activity on economic growth varies from country to country at different stages of development. In the case of developed countries, a positive relationship was found. However, this link is not sustained for nations that are at the initial stages of development.

These paradoxical findings have led to the need to move the analysis toward the various types of entrepreneurial activities that arise in an economy (Baumol, 1990). Implicit in this argument is the fact that not all types of entrepreneurship are associated with economic development (Boettke and Coyne, 2003; Jia, 2018) and more specifically with employment, innovation and welfare (Acs *et al.*, 2008). Therefore, the ways in which diverse societies assign entrepreneurship can restrict or promote economic growth (Williamson, 2013). For example, statistics coming from the Global Entrepreneurship Monitor (GEM) show that Latin American countries exhibit one of the highest rates of entrepreneurial activity. However, that does not correspond to their ordinary rates of economic growth (Larroulet and Couyoumdjian, 2009).

Accordingly, as noted by Desai (2011), in developing countries there are different types of entrepreneurial activities which, in turn, impact economic growth differently. For instance, formal entrepreneurship has been associated with economic development (De Soto, 1989) and there have been very few studied in the literature of entrepreneurship (Dau and Cuervo-Cazurra, 2014; Autio and Fu, 2015). Formal entrepreneurs contribute to the provisioning of goods and public services and social welfare through the payment of taxes and their compliance with environmental and labor regulations. Furthermore, economies with a majority of their activities developed within the formal sector minimize the levels of corruption in the government and therefore contribute to better institutional quality (Dreher and Schneider, 2010). Finally, a vibrant formal sector is a reflection of a level playing field for all economic agents. Through it, entrepreneurs can obtain a productivity benefit from access to public services, such as access to courts to resolve commercial disputes or enforce their property rights (La Porta and Shleifer, 2008; Klapper *et al.*, 2010). These same elements are associated with the development of impersonal markets, and therefore with the growth of enterprises (North, 1990).

However, in the literature the categorization of entrepreneurial activities within the formal–informal sector has not been ruled out of controversy, joined with the difficulty of empirically implementing each of the definitions that have been proposed (Feige, 1990). Despite this ambiguity in the definition, two approaches have been dominant in the literature on formality-informality: "dual" and "legalistic" views (La Porta and Shleifer, 2008, 2014).

The "dual" view is related to the dual economy theory of development (Lewis, 1954). According to this view, informal firms look very different from formal firms. For instance,

"informal firms are those small-scale, low productivity, and using simple production methods and technologies, whereas 'formal' firms are run by much better-educated managers than informal ones, more capital is used and they have different customers, market their products, and use more external finance" (La Porta and Shleifer, 2014, p. 1). At the empirical level, the distinction between formal and informal entrepreneurial activity is determined by labor status: own-account workers vs business owners (Gasparini and Tornarolli, 2009).

The second view in the literature is the "legalistic," which is related to the conceptual framework proposed by The New Institutional Economics (Feige, 1990). According to this view, an economic agent is regarded as a member of the "formal" sector if it adheres to the established rules. At the empirical level, the distinction between formal and informal entrepreneurial activity is determined by registration status (Desai, 2011). Additionally, as a robustness check, we also used a measure of informal entrepreneurial activity: the percentage of the active workforce that is an own-account worker. The purpose was to conduct a comparative analysis allowing determining if institutional factors had a different impact on informal and formal entrepreneurial activity under the two definitions. In the next section, what each measure is capturing will be discussed more deeply, including the limitations and its strengths and weaknesses that should be taken into account in the analysis.

The paper is built on the works of North (1990) and Baumol (1990). The purpose of this research, therefore, is to examine the impact of institutional factors such as business regulation and rule of law over time and on formal entrepreneurship in Latin American countries using two alternative definitions drawn from two databases: The Latinobarómetro database and The World Bank Entrepreneurship Survey (WBGES). These databases allow high comparability at the level of Latin America for several periods of time.

The paper provides several contributions. First, we analyzed together two alternatives views of formal entrepreneurial activity suggested in the literature: the "dual" and "legalistic" view. Second, we contribute to the entrepreneurship literature by analyzing the influence of institutional factors across countries and over time. An analysis of institutional change over time is relevant, especially for developing countries. In the words of Stenholm *et al.* (2013, p. 190) "The results related to high – income countries are likely to remain relatively stable over time, but the analysis of the institutional changes in developing countries seems to guarantee further investigation." For this reason, the time dimension becomes relevant. Third, the lack of studies on types of entrepreneurial activities in developing countries has been happening mainly due to the lack of comparable cross-country data. An unusual data set was used since it could be useful for longitudinal research on entrepreneurial activity in Latin American countries.

Using panel regression techniques, the results suggest that business owners' rate is more substantial in Latin American countries that have weak property rights. Conversely, from a legal definition, formal entrepreneurial activity is more significant in Latin American countries that have most secure property rights and fewer labor regulations. In general, the empirical evidence shows different findings for each definition and those differences are therefore very important to take into account, especially from public policy point of view. We believe that this analysis could be valuable mainly for policymakers in Latin American countries because it will allow them to understand the institutional factors that underlie the decisions of entrepreneurs to operate in the formal sector.

The article is organized as follows: Section 2 discusses theory and hypothesis development. Section 3 describes the variables and econometric method used in the empirical analysis. In Section 4, the statistical analysis is developed. Then, in Section 5 the empirical results are presented. Finally, in Section 6 we conclude and discuss future research lines.

2. Theory and hypothesis development

2.1 Defining formal entrepreneurial activity

Defining and measuring formal–informal activity is a complicated issue since the concept of "Informality" means different things to different people (Gasparini and Tornarolli, 2009). Therefore, there exists a limited consensus in the academic literature on how to define and operationalize these concepts in developing economies (Feige, 1990). For example, for Kanbur (2009, p. 2) "Informality is a term that has the dubious distinction of maximum importance and political salience with minimal policy combining conceptual clarity and coherence in the analytical literature."

Despite this ambiguity in the definition, two approaches have been dominant in the literature: the "dual" and the "legal" view (La Porta and Shleifer, 2008). The dual approach is associated with the dual view of development (Lewis, 1954). According to this approach, the informal economy is related to those working in low productivity, unskilled, marginal jobs and with low capacity of accumulation (La Porta and Shleifer, 2008; Gasparini and Tornarolli, 2009). At the empirical level, specific labor categories, such as unskilled own-account workers, are more likely to form part of such activities (Loayza and Rigolini, 2011).

For example, Aguilar *et al.* (2013), using information from the Latinobarómetro database, found that own-account workers enjoy a lower level of work satisfaction than business owners. Also, Mondragón-Vélez and Peña (2010, p. 124), using information from household surveys, conclude "that while business ownership shares the main characteristics of what the literature associates with entrepreneurship, self-employment in the Colombian context is more associated with a subsistence activity." Moreover, self-employed workers are more likely to hide their productive activities from the tax and registry authorities (Portes and Haller, 2005). In the study mentioned above, Mondragón-Vélez and Peña (2010) found that only five percent of own-account workers registered their activity in Colombia over the period 2002–2006.

Therefore, in Latin American countries, the self-employed have less human capital, smaller incomes and lower levels of satisfaction than business owners (Mondragón-Vélez and Peña, 2010; Aguilar *et al.*, 2013). These findings contradict the traditional models of occupational choice that have been developed primarily to understand the labor dynamics of developed countries. In these models, the decision of individuals to become entrepreneurs depends, among other things, upon the higher returns that they expect to obtain relative to wage employment (Lucas, 1978) and the preference of certain individuals for greater autonomy (Blanchflower and Oswald, 1998).

From these findings, differentiating between own-account workers and businesses owners in developing countries seems quite relevant (Mondragón-Vélez and Peña, 2010). Also, the fact that business owners have employees in their charge may be a good sign of a successful business which was started by opportunity, different from the marginal and low productivity jobs that characterize the own-account workers. For instance, Gindling and Newhouse (2014) classify entrepreneurs as successful whether the self-employed are employers (vs own-account workers). Based on these findings, in this paper, business owners are considered a reflection of formal entrepreneurs.

Regarding the "legal" definition of formal–informal activity, it is much more accurate at a theoretical level than the dual definition. However, it is more challenging to implement at the practical level (Gasparini and Tornarolli, 2009). Under the new institutional conceptual framework, Feige (1990, p. 5) argues that "an economic agent is regarded as a member of the 'formal' sector of any economy when his actions adhere to or are protected by the established institutional rules of the game." Therefore, the distinction between formal and informal entrepreneurial activity is determined by registration status (Desai, 2011). The above means that inclusion in the formal sector is not related to the nature of the business or

labor status, but is connected with the non-compliance of regulations implemented by the State (De Soto, 1989). The legal definition has been extensively used by scholars in the field of entrepreneurship, while the dual definition has been used primarily by those in the field of Labor Economics.

2.2 Formal institutions and formal entrepreneurial activity

Formal institutions are those which are officially codified in written documents (Lauth, 2015). For North (1990), formal rules include political rules, economic rules and contracts. Formal institutions are mainly related to the establishment of property rights and enforcement of contracts (North, 1990). Empirical works carried out in recent decades by authors such as Knack and Keefer (1995) and Acemoglu *et al.* (2001), among others, have confirmed the positive impact of institutions on economic development. At the same time, the empirical results suggest that institutional factors dominate other economic factors that have traditionally been considered as the fundamental causes of economic growth, such as investment and human capital (Romer, 1986) or factors such as geography and even from an older tradition such as culture (Landes, 1998).

If we follow the general definition of institutions from North (1990), institutions are the rules of the game in society. The evidence in the field of comparative economics suggests a diversity of legal frameworks among countries (Djankov *et al.*, 2003), determined in part by its legal tradition (La Porta *et al.*, 1998). Therefore, each legal framework will produce certain types of economic incentives. Some of these incentives will be aligned with the kind of entrepreneurial activity consistent with economic growth while others will not be (Boettke and Coyne, 2003; Stenholm *et al.*, 2013).

Property rights have been identified as a critical element that captures the essence of the quality of economic institutions. In those societies where property rights are well defined and contracts are enforced, individuals will have incentives to invest in physical or human capital or adopt improved technologies (Acemoglu *et al.*, 2005). Likewise, in societies with a firm rule of law, the incentives that individuals obtain for participating in the formal sector, such as access to courts to resolve commercial disputes or the availability of credit in the official financial market, will offset the various costs associated with formality such as taxation or labor regulation (La Porta and Shleifer, 2008; Desai, 2011). On the other hand, in societies with a weak enforcement of contracts it is more likely that entrepreneurs operate outside the radar of the government authorities (Almeida and Carneiro, 2012).

In the field of entrepreneurship, there is a relevant body of literature, which analyzed the relationship between property rights and different types of entrepreneurial activities at the empirical level. For instance, Williamson (2013) and Simón-Moya *et al.* (2014) found that the degree of economic freedom is related to high levels of opportunity entrepreneurship. The authors built an Index of Economic Freedom from Heritage Foundation, which includes property rights (among other indicators). In the same vein, Nyström (2008) uses an array of similar predictors. The author used COMPENDIA database covering 23 OECD countries. The results suggested that better legal structure and security of property rights tend to increase entrepreneurship measured as self-employment rates. Meanwhile, McMullen *et al.* (2008) and Estrin *et al.* (2013), using data provided by the GEM, also found that opportunity and high growth entrepreneurship is positively related to property rights.

Concerning formal entrepreneurship, most studies have used The WBGES. This database collects information from several countries around the world on business registered in a public registry. For example, Klapper *et al.* (2011) found a positive relationship between formal entrepreneurship and a measure of national governance comes from the World Bank Worldwide Governance Indicators, which include the rule of law variable.

This same relationship was found by Thai and Turkina (2014) and Dau and Cuervo-Cazurra (2014). Meanwhile, Autio and Fu (2015) found that the quality of political institutions measured by the Political Rights index from Freedom House has a positive relationship with formal entrepreneurship. According to Acemoglu *et al.* (2005), the quality of political institutions is a good predictor of the quality of economic institutions, because the latter are endogenous since the political system creates them.

In summary, the presence of secure property rights improves the exploitation of entrepreneurial opportunities, because this reduces the risk of expropriation from the Government and elites (Acemoglu *et al.*, 2005) and the transaction costs in the economy (North, 1990). For instance, the existence of an impartial justice system facilitates the resolution of commercial disputes and minimizes the risks of opportunism in business (La Porta and Shleifer, 2008). The above, in turn, creates incentives for a higher number of people to engage in economic transactions, and furthermore, they can make long-term investments, mainly in high-productivity sectors such as industry (Knack and Keefer, 1995). According to Klapper *et al.* (2010), the industry is one kind of activity more inclined to formality. Therefore, based on these findings, we propose the following hypothesis:

H1. Stronger property rights are associated with higher levels of formal entrepreneurial activity.

In his famous book *The Other Path*, De Soto (1989) described the results of the fieldwork carried out by him and his group of researchers from the Institute for Liberty and Democracy regarding the list of required procedures by actually going through the process of setting up a business in Peru. The research determined the existence of an excessive regulatory burden that stifled formal entrepreneurship, forcing many Peruvian entrepreneurs into the informal economy. In turn, this work inspired a group of researchers from the World Bank to replicate the study in several countries around the world. The results were published in the paper of Djankov *et al.* (2002). In this work, the authors found that countries differ significantly in the way in which they regulate the entry of new businesses. From this study, the World Bank has each year been measuring indicators related to the procedures for operating a business, as well as the time and cost of following these procedures.

Based on information from The World Bank's Doing Business Surveys, several papers have analyzed the possible relationship between the regulatory framework and entrepreneurial activity. For instance, La Porta and Shleifer (2008) and Klapper *et al.* (2010) using information on registered firms from The WBGES found that the number of procedures necessary to start a business is associated with fewer registered firms. These same findings were observed by Klapper *et al.* (2006), using information from a private database and focusing on new corporation creation in Western and Eastern Europe. On the other hand, the results of the work of Aparicio *et al.* (2016) ratified this relationship for the case of opportunity entrepreneurship. The authors used an unbalanced panel data with 43 countries for the period 2004–2012. Finally, Levie and Autio (2011) using a panel of six years and 54 countries, found that the burden of regulation, which is based on several indicators from the World Bank's Doing Business, was related to a higher rate and relative prevalence of strategic entrepreneurial entry.

Regarding labor regulation, Scarpetta *et al.* (2002), in a study for the OECD countries, found that labor market regulations are negatively correlated with the number of new small and medium-sized companies. This same relationship was observed by Klapper *et al.* (2006) in labor-intensive industries in Western and Eastern Europe. Meanwhile, Van Stel *et al.* (2007), using data from the GEM for the period 2002–2005 and 39 countries, found that labor market rigidity hurts nascent entrepreneurship rates. In the same vein,

McMullen *et al.* (2008) found a positive relationship between labor flexibility and opportunity entrepreneurship. However, Klapper *et al.* (2010) and Djankov *et al.* (2010) did not find a significant association in the case of newly registered firms from The WBGES database. Therefore, based on these results, we propose the following hypotheses:

H2. The number of days involved in starting a business is associated with lower levels of formal entrepreneurial activity.

H3. Stricter regulation of labor is associated with lower levels of formal entrepreneurial activity.

3. Methodology

3.1 Data and variables

As previously established, this paper seeks to estimate the relationship between business regulation, rule of law (formal rules) and formal entrepreneurship for different Latin American countries over several time periods. For this purpose, we have built two panels. The first panel covers the period 2004–2015, and the data set contains 180 observations in 18 countries: Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Dominican Republic, Ecuador, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Paraguay, Peru, Uruguay and Venezuela. The second panel covers the period 2006–2015, and the data set contains 134 observations in 14 countries: Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Dominican Republic, El Salvador, Guatemala, Mexico, Panama, Paraguay, Peru and Uruguay.

3.1.1 Dependent variables. The database used for the analysis of the "legal" definition of formal entrepreneurial activity comes from The WBGES. The survey has been designed to be comparable across countries and formally measures entrepreneurship as the number of new officially registered limited liability corporations (LLCs) (Klapper *et al.*, 2010). That is to say that it only includes enterprises incorporated as legal entities and registered in a public registry. Specifically, the dependent variable that we used to measure the level of formal entrepreneurial activity from the legal definition corresponds to the business entry density rate, which is defined as the number of newly registered firms with limited liability per 1,000 working-age people (ages 15–64) per the calendar year. Among the limitations of the data, it is questionable to assume that only LLCs adhere to prevailing legal regulations.

On the other hand, the data used for the analysis of the "dual" definition of formal entrepreneurial activity comes from Latinobarómetro. The surveys have been conducted in the region since 1995 until the present date; however, since 2004, the surveys have presented national coverage of nearly one hundred percent for all countries and a standard questionnaire leading to harmonized data. The ten measurement surveys consist of approximately 1,000–1,200 surveys per country. The samples are representative of the adult population of each country, with a margin of error of approximately 3 percent. The design and focus of the surveys are comparable to that of the Eurobarometer survey for European countries.

As mentioned above, there are several empirical studies carried out in developing countries and Latin America in particular that suggest that significant differences exist in terms of human capital, incomes, work satisfaction and traits psychological associated with entrepreneurs, between own-account workers and business owners (see Mondragón-Vélez and Peña, 2010; Aguilar *et al.*, 2013; CAF, 2013; among others). Therefore, a characterization of entrepreneurial activities is relevant. Also, the own-account workers are more associated with subsistence and very low productivity. Furthermore, they have typically been associated with the informal sector (Hart, 1973), given that in developing countries, a strong

association has been found between self-employment and informal activity (La Porta and Shleifer, 2008; Loayza and Rigolini, 2011).

Therefore, the dependent variable that we use to measure the level of formal entrepreneurial activity from the "dual" view corresponds to the percentage of the adult population who are identified as owners of a non-agricultural business. The use of business ownership as a measure of entrepreneurship has been pointed out by Van Stel (2005). Nevertheless, the inclusion of business owners as formal entrepreneurs is debatable, mainly because it includes micro-entrepreneurs (i.e. fewer than five employees) who also have been identified with enterprises of low productivity, low capital, and which do not comply with current regulations (Lederman *et al.*, 2013).

Finally, it should be noted that the "dual" measure of formal entrepreneurial activity reflects the stock of business owners in a given year. Therefore, it is a static measure of entrepreneurial activity. On the contrary, the "legal" measure of formal entrepreneurial activity reflects the entry of new firms registered in the economy. Therefore, it is a measure closest to the definition of entrepreneurship.

3.1.2 Independent variables. As independent variables, we included regulatory business and property rights. The regulatory business includes two variables: business entry cost (World Bank Doing Business Project) and which reflect the number of days to start a new business. The other variable corresponds to obeying government regulations such as labor regulations (World Bank Doing Business Project). To this end, we built the Rigidity of Employment index: a simple average of three sub-indices: Difficulty of Hiring Index, Rigidity of Hours Index and Difficulty of Firing Index (World Bank Doing Business Project). The index was constructed following Berg and Cazes (2007), based on the work of Botero *et al.* (2004).

Moreover, as a measure of Property rights, we used the Rule of Law Index (World Bank's Governance Indicators project). According to Kaufmann *et al.* (2011, p. 223) this index capture "perceptions of the extent to which agents have confidence in and abide by the rules of society, and in particular the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence."

3.1.3 Control variables. Several studies have found a relationship between the level of entrepreneurial activity and economic development (Van Stel *et al.*, 2005; Levie and Autio, 2011; Estrin *et al.*, 2013). For this reason, we have used GDP per capita (purchasing power parity) as a control variable. Similarly, we have used the economic growth rate (Dau and Cuervo-Cazurra, 2014), and population growth (Levie and Autio, 2011; Autio and Fu, 2015) as control variables. It is reasonable to expect that to an economy with a higher population, and economic growth rate should present better entrepreneurial opportunities for formal entrepreneurial activity. Additionally, formal entrepreneurial activity is related to the educational level and the unemployment rate (Thai and Turkina, 2014). As a control for education level, we used the average years of secondary schooling of the adult population (Barro and Lee, 2013). Here, we include time-fixed effects to capture the business cycle of these countries.

4. Statistical analysis

To analyze the relationship between formal rules (business regulation and rule of law) and the two definitions of formal entrepreneurship in Latin America, we used two data panel settings. The first corresponds to a balanced panel of 18 countries, covering the period 2004–2015. The second is an unbalanced panel of 14 countries during the period 2006–2015. Therefore, we considered two models: model one shows the relationship between formal rules and formal entrepreneurship from a legal definition. Instead, the two

shows the relationship between formal rules and formal entrepreneurship from a dual definition. We have estimated the following equations:

$$\text{Formal Entrepreneurial Activity(legal definition)} = \beta 0 + \beta 1.GDP_\text{percapita} + \beta 2.\text{Growth_rate}_GDP + \beta 3.\text{Population_Growth} + \beta 4.\text{Secundary_Education} + \beta 5.\text{Unemployment} + \beta 6.\text{Business_entry_cost} + \beta 7.\text{Rigid_labour} + \beta 8.\text{Rule of law} + \beta 9.\text{Trend}(t) + u. \quad (1)$$

$$\text{Formal Entrepreneurial Activity(dual definition)} = \beta 0 + \beta 1.GDP_\text{percapita} + \beta 2.\text{Growth_rate}_GDP + \beta 3.\text{Population_Growth} + \beta 4.\text{Secundary_Education} + \beta 5.\text{Unemployment} + \beta 6.\text{Business_entry_cost} + \beta 7.\text{Rigid_labour} + \beta 8.\text{Rule of law} + \beta 9.\text{Trend}(t) + u. \quad (2)$$

First, we verified whether the use of panel data vs a simple OLS regression was justified. This was done by applying the Breusch and Pagan test. We rejected the null hypothesis (Prob > chibar2 = 0.0000 in both models), which states that panel data are preferable to using a pooled data set. To verify whether it was preferable to use fixed effects or random effects, we applied the Haussmann test. The test suggested that the random effects model was better suited to both models (p-value = 0.27 for model 1 and p-value = 0.12 for model 3). We also verified the presence of autocorrelation. This was done by applying the test for serial correlation derived by Wooldridge (2002). We did not reject null hypothesis for the models 1 and 3 (p-value = 0.81 for model 1 and p-value = 0.79 for model 3). Therefore, we re-estimated the equations through random effects generalized least squares (GLS) with panel-specific autocorrelation AR(1). Finally, we verified the existence of severe multicollinearity problems, mainly taking into account that the correlation matrix (Table II) included some correlations over 0.5. To this end, we applied the variance inflation factor; which in any case shows values below the commonly accepted threshold of 5 and 10 (Dau and Cuervo-Cazurra, 2014). Therefore, multicollinearity did not appear to be a problem.

5. Results

Table I shows the means and standard deviations for the variables analyzed as well as their descriptions and sources from where they were taken. The model 1 in Table II presents the results based on the effect of institutional and control variables on formal business registration (legal definition). *H1* and *H3* were supported. Stronger rule of law are associated with higher rate of business registration ($p < 0.01$). This result is consistent with Klapper *et al.* (2011) and Stenholm *et al.* (2013) results. Besides, stricter labor regulation had an adverse effect on rate of business registration ($p < 0.01$). This result differs from those obtained by Klapper *et al.* (2010) and Djankov *et al.* (2010) who did not find any statistically significant effect on formal entrepreneurship using The WBGES database.

In summary, an improvement in Rule of law by one standard deviation results in an increase in the formal business registration rate of about 0.275 standard deviations; while stricter labor regulations by one standard deviation results in a decrease in the business registration rate of 0.192 standard deviations. Furthermore, we failed to find any effects of business entry on formal business registration rate. The entrepreneurs of registered firms likely pursue more valuable entrepreneurial opportunities and are willing to bear entry costs (La Porta and Shleifer, 2008).

Moreover, model 3 in Table II presents the results based on the effect of institutional and control variables over business owners' rate (dual definition). No hypothesis was supported

Variable	Measure	Description	Mean	SD
1. Formal entrepreneurial activity Rate (dual definition)	Rate	Rate of the adult population who identify themselves as owners of the non-agricultural business (Latinobarómetro)	6.87	3.01
2. Formal entrepreneurial activity Rate (Legal definition)	Rate	The number of newly registered firms with limited liability per 1,000 working-age people (ages 15–64) per the calendar year (World Bank Group Entrepreneurship Survey)	1.45	1.52
3. GDP per capita	USD	GDP per capita, purchasing power parities. (log for analysis) (International Monetary Fund Data)	11,220.52	5,232.46
4. Growth rate GDP	Percent	Growth rate GDP (International Monetary Fund Data)	4.42	3.24
5. Population growth	Rate	The population growth rate (International Monetary Fund Data)	1.42	0.56
6. Secondary education attainment	Percent	Average Years of Secondary Schooling (Barro-Lee Educational Attainment Data – World Bank Data Base)	2.40	0.72
7. Unemployment rate	Percent	Percent of total labor force (World Bank Data Base)	6.71	2.42
8. Business entry cost	Number	The number of days to start a new business (log for analysis) (Doing Business)	48.90	39.36
9. Rigidity of employment	Index	It is a quantitative measure that considers aspects of the regulatory framework of a country's labor market Normalized between 0 and 1. A higher value indicates greater rigidity (Doing Business)	0.40	0.18
10. Rule of law	Index	This index measures "[…] the quality of contract enforcement, the police, and the courts, and also the likelihood of crime and violence" (Kaufmann *et al.*, 2011). Normalized between 0 and 1. A higher value indicates stronger property rights (World Bank Governance Indicators)	0.37	0.16

Table I.
Description of variables and statistics

by the dual definition of formality. Contrary to *H1*, rule of law had a negative and strongly significant effect on business owners' rate. One possible explanation for this finding is the presence of small-scale business owners (less than five employees). Therefore, the results suggest that stronger property rights will lead to a smaller number of micro-entrepreneurs. Hence, our dual definition of formal entrepreneurial activity does not seem to capture the type of entrepreneurial activity positively related to higher economic dynamism.

In the Table II (model 5), we have included results for the case of informal entrepreneurial activity. For this, we use a measure of informal entrepreneurial activity. This measure comes from ILOSTAT database and corresponds to the percentage of the active workforce that is an own-account worker[1]. The purpose was to conduct a comparative analysis allowing determining if institutional factors had a different impact on informal and formal entrepreneurial activity under the two definitions.

First, if we compare informal and formal entrepreneurial activity by the legal definition (see Table II, models 1 and 5), the results are reasonably consistent with theory and previous empirical evidence That is to say, there is strong support for the argument that institutional factors impact different formal and informal entrepreneurial activities, as suggested by several works (Dau and Cuervo-Cazurra, 2014; Thai and Turkina, 2014; Autio and Fu, 2015). For example, a better rule of law had a positive effect on the entry of new firms registered, but an adverse effect on the level of informal entrepreneurial activity. Moreover, while

	Legal definition Robustness checks		Dual definition Robustness checks		Informality
	Model 1 (GLS)	Model 2 (GLS)	Model 3 (GLS)	Model 4 (GLS)	Model 5 (GLS)
Business entry cost	0.00175 (0.0771)	0.0656 (0.0783)	−0.000734 (0.00190)	−0.00102 (0.00180)	−0.000504 (0.0102)
Rigidity employment	−1.688** (0.555)		−0.0107 (0.0100)	−0.0129 (0.0109)	−0.0651 (0.0590)
Rule of law	3.082** (1.156)		−0.0841*** (0.0112)		−0.665*** (0.0840)
Labor freedom		1.747*** (0.482)			
Property rights		1.655** (0.574)		−0.0255*** (0.00702)	
Constant	−1.772 (2.254)	−0.543 (2.014)	0.162*** (0.0371)	0.194*** (0.0384)	6.073*** (0.236)
Observations	122	134	162	180	198
Groups (max.)	9	10	18	18	18
Groups (aver.)	8.714	9.571	9	10	11
Groups (min.)	6	6	9	10	11
Wooldridge test (p-value)	0.89	0.63	0.81	0.83	0.72
Hausman test (p-value)	0.27	0.00	0.12	0.01	0.18
Time effects	Yes	Yes	Yes	Yes	Yes
χ^2	26.88	67.71	136.2	130.1	285.6

Notes: Estimators used: Generalized least squares (GLS). All coefficients are unstandardized. Standard errors in parentheses. The controls included in all models are: GDP per capita; population growth; GDP growth rate, unemployment and secondary education level, but results are not reported for space reasons. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

Table II.
Formal entrepreneurial activity rate (Legal and dual definition)

business entry was not significant, the sign of the coefficient was negative for informal activity and positive for formal business registration rate.

Additionally, we made a comparison between informal and formal entrepreneurial activity by the dual definition (see Table II, models 3 and 5). In general, the results suggested that formal rules impact both types of activities in the same direction. For instance, stronger property rights will lead to less own-account workers and business owners. This counter-intuitive result would suggest that a better institutional framework not only would lead to a less amount of own-account workers, but also to a less amount of small-scale business owners. These paradoxical results seem to be consistent with several studies that have pointed out that one of the most serious problems of productivity and competitiveness of Latin American enterprises is their size and lack of capacity to grow (Lederman *et al.*, 2013).

Therefore, the results support the importance of conducting analyzes that take into account the different types of entrepreneurial activities that are present in an economy, and in addition the relevance of understanding what each measure is capturing of the heterogeneous phenomenon of entrepreneurship. It is important to point out that rule of law was a robust variable for all our models and specifications. The Stronger rule of law (property rights) leads to lower own accounts workers and business owners (dual definition) and a higher business registration rate (legal definition).

5.1 Robustness checks

As a verification of the robustness of our results in the Model 1, we also employed two alternative measures for rule of law and rigid labor (model 2). As an alternative proxy for rule of law, we used property rights index from Heritage Foundation, also as an alternative measure to rigid labor, labor freedom from Heritage Foundation was used. As displayed in the model 2 (Table II), the results are maintained and even increase their significance. As a verification of the robustness of our results in the model 3, we again employed Property Rights Index from Heritage Foundation as one alternative measures for rule of law. As displayed in the model 4 the results are similar to the Model 3. Therefore, the results proved robust against specification changes.

Additionally, we applied others robustness checks which were not presented to save space. For instance, we used an appropriate alternative method for data panels: a time-series generalized estimating equation (GEE) model (Klapper *et al.*, 2010; Dau and Cuervo-Cazurra, 2014). The results were very similar to those obtained through random effects. Therefore, the results are robust to the estimation method used (RE or GEE). Finally, we lag each of the independent variables by one year. In both models, the results were maintained. Overall, the results corroborated the robustness of our previous findings.

6. Discussion and conclusions

In this work, we have empirically analyzed the relationship between business regulation, rule of law and formal entrepreneurship in Latin American countries based on institutional theory (North, 1990). This was done using two panels that capture two alternatives definitions of formal entrepreneurship suggested in the literature: dual and legal definition. The first panel covered the period 2004–2015 with a data set containing 180 observations in 18 Latin America countries, while the second panel covered the period 2006–2015 with a data set containing 134 observations in 14 Latin America countries.

In summary, the impact of institutional variables on formal entrepreneurship depends on the definition employed. In the case of the dual definition, the results suggest that business owners' rate is more substantial in Latin American countries that have weak property rights. Conversely, from a legal definition, the business registration rate is more significant in Latin American countries that have the most secure property rights and fewer labor

regulations. These contradictory results suggest that the legal definition of formal entrepreneurship, except for the dual definition, seems to be associated with the type of entrepreneurial activity that promotes economic growth and development. The results were maintained after being controlled by some socio-economic factors, such as GDP per capita, the population growth rate, GDP growth rate, unemployment rate and education level.

The results obtained in the present study, through legal and dual definition of formal entrepreneurship, contribute to show the existence of different types of entrepreneurial activities. For example, when the dual definition, which included micro-entrepreneurs, was applied, the findings showed a strong and negative correlation between property rights and business owners. That means that better property rights not only leads to a lower amount of own-account workers (Salinas *et al.*, 2018) but also business owners and more specifically a smaller number of small-scale business owners. However, the results also show that better property rights are associated with an increased entry of registered firms (the legal definition of formal entrepreneurship), which are related to types of entrepreneurial activity with higher expectations of growth (Lederman *et al.*, 2013). For instance, Klapper *et al.* (2011) found that the number of registered firms is higher in developed than in developing countries. In summary, these two definitions are capturing different dynamics of entrepreneurship. More concretely, the measure used for the legal definition reflects quality of entrepreneurial activity, whereas, the dual definition captures the quantity of entrepreneurial activity.

Aligned with what was stated above, the results help to shed some light on the debate that exists in the literature between the dual view of informality and the position related to Hernando De Soto. From De Soto's viewpoint, if governments reduce their regulatory burden and improve the legal framework, informal entrepreneurs will channel their entrepreneurial energies to the formal sector, generating a virtuous cycle of employment and prosperity. Furthermore, from the dual view of informality, their advocates do not deny the importance of the regulatory and legal framework. However, they consider that those institutional hindrances, rather than constraining informal entrepreneurs, are imposing high entry barriers to the formal sector on those potential entrepreneurs with more considerable human capital, entrepreneurial talent and technological capacity to establish enterprises with high growth potential. In turn, those new firms will absorb the millions of own-account workers who currently operate in the economies of developing countries. Therefore, if the perspective of De Soto is correct, as a result of an improvement in the rule of law, we should observe a reduction in the number of self-employed workers and an increase in the business ownership rate. Nonetheless, the findings do not suggest a transition between self-employment and business ownership. Instead, the data reveal an increase in the number of limited liability companies, which are related to people with more considerable human capital, entrepreneurial talent and technological capacity. Therefore, our findings favor the dual view of informality.

Also, the results provide some support for the U-shaped relationship found between economic development and the national rate of entrepreneurship (Van Stel *et al.*, 2005). The evidence has suggested that as a country develops, the rate of entrepreneurial activity will tend to fall; however, at a high level of development, there will be a reversal of the trend. Our findings confirm that at a low-development stage, the quantity of own-account workers and business owners is high. Yet to the extent that nations improve the regulatory and legal framework, both, the number of own-account workers and business owners tend to fall. At the same time, the number of LLCs increase. In other words, to the extent that the countries move toward higher development stages, the quantity of entrepreneurship will decline, whereas the quality of entrepreneurship increases.

A relevant contribution of this research, useful as far as public policy is concerned, is that it provides evidence of the existence of different types of entrepreneurial activities, which in

turn respond differentially to the incentives arising from the institutional framework. Therefore, the primary policy implication is that each of these entrepreneurial activities requires different types of public policy. For instance, aligned with the dual vision of informality, the public policy, which is most efficient to reduce high levels of informality, is related to a one growth strategy for formal firms and incentives for the creation of highly productive formal firms. In both cases, the aim is to create quality employment that enables the absorption of informal and micro-entrepreneurs.

For example, the evidence on entrepreneurial demography in Latin American shows the existence of a large number of micro-enterprises, many of them unproductive. However, at the same time, there are a small number of medium-sized enterprises. This fact is primarily associated with the low productivity of micro-enterprises that do not allow them to transit toward larger, dynamic entrepreneurial structures. Therefore, public policy must address the factors that underlie this lack of growth. For example, as shown by the research, a key factor is the rule of law. In countries with a better rule of law, employers have a greater incentive to invest. On the other hand, a firm rule of law reduces the level of uncertainty, which in turn encourages entrepreneurs to move their investments from less risky sectors like trade toward areas of higher risk and productivity such as industry, where it is necessary to achieve economies of scale. Also, industry is one of the sectors with a higher capacity to absorb unskilled labor, which is precisely the factor that abounds in the informal sector.

The other channel to reduce the high levels of informal entrepreneurial activity is related to quality of entrepreneurship. According to the dual view of informality, the reduction of the informal economy will depend to no small extent on the creation of new businesses that are legally registered and run by new and better-educated managers (La Porta and Shleifer, 2008). Hence, public policy should create the right incentives to achieve a better allocation of entrepreneurial talent. The present work has shown that the rule of law is a critical element for the creation of new businesses that are legally registered in the form of Limited Liability Company. Therefore, the entrepreneurship policy should focus on the quality of entrepreneurship, rather than merely seeking to increase the number of new businesses.

Additionally, the results can contribute to the discussion that exists in civil society concerning whether governments and the general population should be permissive with informal entrepreneurial activity; or whether, by contrast, it is necessary to force them to comply with government regulations. The latter option seems to be unsuitable. For instance, our results suggest that the reduction of informality depends more on the entry of new firms registered in the form of Limited Liability Company. In other words, in the absence of improvement in the rule of law and regulatory burden, which encourages the business registration rate, probably an aggressive policy forcing the informal entrepreneurs to comply with legal regulations will lead to higher unemployment and poverty, taking into account that around 50 percent of the economically active population in Latin America participate directly or indirectly in the informal economy. Therefore, civil society must take into account all these elements to promote and contribute to a positive and realistic debate over informality.

In summary, we suggest that the objective of entrepreneurial policy in Latin America should focus on the quality of entrepreneurship rather than the number of entrepreneurs. Nonetheless, while Latin America countries manage to stimulate the number of registered firms, informal entrepreneurial activity is essential as a source of employment that provides a livelihood for millions of economically disadvantaged individuals. Therefore, the improvement of property rights and the regulatory burden should aim, as its primary objective, to reward and encourage individuals with the best human capital and entrepreneurial talent to become entrepreneurs. In turn, they will generate quality employment, absorbing millions of informal

entrepreneurs currently inventing jobs to escape poverty. Additionally, the results suggest that the legalistic definition of formal entrepreneurship is probably the most relevant for many policy issues.

Finally, we must acknowledge the limitations of the work. For instance, the work has not explicitly addressed the possible, or not, substitution between formal and informal entrepreneurial activity. However, a joint analysis of all the results suggests that better institutional quality will lead not only to a smaller quantity of own-account workers (informal entrepreneurship) but also to a lower number of business owners (formal entrepreneurship-dual definition). Therefore, it is difficult to think that merely by improving the legal and regulatory framework there will be a massive conversion of informal entrepreneurs to official status. Nevertheless, future research on this topic is necessary.

Note

1. Loayza and Rigolini (2011) suggest using self-employment as a good proxy for informality in developing countries.

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